

Predictions of Small-Body Physical Traits Based on Orbital Characteristics

[Small Body Project Repo](#)

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- **Brainstorming & Dataset Selection (No Citation Required):** **Google Gemini** and **OpenAI ChatGPT** were utilized to brainstorm and isolate a viable machine learning regression scope project for the author selected [JPL NASA Small-Body Dataset](#).
- **Data Selection & Filtering (Informed by AI):** **Google Gemini** acted as a technical guide to filter query parameters on the [JPL NASA Small-Body Dataset](#) web interface. Based on these logical constraints, a custom CSV file was generated and downloaded manually, which is preserved in the project's GitHub repository.
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Problem Statement & Hypothesis Formulation

I. Problem Statement

State the occurrence that you are attempting describe using data science

Two ways to predict the **diameter** of a small-body are **(1)** using a combination of **very high data collection cost**, **very high predictive power**, and **low observational reliability** features like **albedo** (Group 3) with **low data collection cost**, **medium observational reliability** features, like **absolute magnitude (H)** (Group 2) and **(2)** using a combination of **low data collection cost**, **high observational reliability** features (Group 1) with **low data collection cost**, **medium observational reliability** features, like **absolute magnitude (H)** (Group 2)(Ge, 2026; Murcia, 2023; Robinson, 2024). The project looks to determine if the **diameter** of a small-body can accurately be predicted with only Group 1 features to maintain the low-cost, high-reliability of features that are inexpensive and not variable or prone to error while observing.

II. Hypothesis Formulation

Suggested explanation to the problem statement.

The project utilizes a two-model comparison framework that will be assessed from a statistical and operational perspective. The regression models `target diameter (Y)` using two feature subsets: an expanded set of `low-cost, high-reliability` observations with `low-cost, medium-reliability` observations ($\mathbf{X}_{full}$) and a restricted set of only 'low-cost', 'high reliability' observations ($\mathbf{X}_{restricted}$). The restricted model ($\mathbf{X}_{restricted}$) is preferred due to the lower number and higher reliability of features. Current peer-reviewed research dictates that `absolute magnitude (H)`, a Group 2 feature, is mathematically required to calculate `diameter` (Ge, 2026).

The hypothesis determines if small-body `diameter (Y)` can be accurately predicted within a statistical and operational threshold (δ) by the restrictive feature subset compared to the full feature subset, quantified as:

$$\Delta R^2_{test} = R^2_{Model(\mathbf{X}_{full})} - R^2_{Model(\mathbf{X}_{restricted})}$$

Statistical Hypothesis

The statistical hypothesis tests if the restrictive feature subset of Group 1 features maintains a statistically acceptable degree of accuracy in predicting `target diameter (Y)` relative to the expanded feature subset of Group 1 and Group 2 features.

- **Statistical Null Hypothesis ($H_{0,stat}$):** $\Delta R^2_{test} \leq 0$

The inclusion of Group 2 features yield no improvement or a decrease in the out-of-sample ΔR^2_{test} , indicating that the expanded features **do not** increase the accuracy of predicting the `target diameter (Y)`.

- **Statistical Alternative Hypothesis ($H_{1,stat}$):** $\Delta R^2_{test} > 0$

The inclusion of Group 2 features yields a statistically significant increase in the out-of-sample ΔR^2_{test} , indicating that the expanded features **do** increase the accuracy of predicting the `target diameter (Y)`.

Operational Hypothesis

The operational hypothesis tests if the restrictive feature subset of Group 1 features maintains an operationally and practically acceptable degree of accuracy (δ) in predicting `target diameter (Y)` relative to the expanded feature subset of Group 1 and Group 2 features.

- **Operational Null Hypothesis ($H_{0,ops}$):** $\Delta R^2_{test} > \delta$

The inclusion of Group 2 features yield an operationally significant increase in accuracy to predict `target diameter (Y)`, indicating that the restricted feature subset **is not** sufficient enough for operational use.

- **Operational Alternative Hypothesis ($H_{1,ops}$):** $\Delta R^2_{test} \leq \delta$

The inclusion of Group 2 features does not yields an operationally significant increase in accuracy to predict `target diameter` (`Y`), indicating that the restricted feature subset **is** sufficient enough for operational use.

The Data Science Process

I. Acquire

Identify data sets, retrieve data, query data.

In order to test the two-model comparison framework, a comprehensive and reliable dataset is required. The dataset needed to contain characteristics from Group 1 and Group 2 for small-bodies. The [JPL NASA Small-Body Dataset](#) was selected because NASA is an authoritative, reliable, and primary repository of small-body data that is also open-source and available to the public allowing reproduction of testing for peer-review.

The [JPL NASA Small-Body Dataset](#) contains Group 1 and Group 2 characteristics such as `absolute magnitude` (`H`), `semi-major axis` (`a`), `eccentricity` (`e`), and `inclination` (`i`) which directly relate to the problem statement and hypothesis requirements necessary for the expanded features subset and restricted feature subset.

The [JPL NASA Small-Body Dataset](#) was queried to meet the `target diameter` (`Y`) to be used in out-of-sample testing. The query exclusively included asteroids with a known diameter greater than zero kilometers (0 km). This ensures a valid target variable for regression training and the calculation of the coefficient of determination (R^2). Based on these parameters, a custom CSV file was generated and downloaded manually, which is preserved in the project's GitHub repository.

II. Prepare

Explore and pre-process. Data has been prepared for regression analysis in the following ways:

1. **Data Structure Integrity:** Validate all data types are correct. Name columns should be strings (`str`), continuous values should be floats (`float64`), and discrete values should be integers (`int`).
2. **Data Domain Validation:** Remove data is implausible such as negative `diameter` .
3. **Duplicate Data Removal:** Remove duplicate rows of data and observations of the same small-body. Multiple observations of the same small-body skew the sampling data in regression and create a data leakage problem with artificial accuracy in the

case where an small-body observation ends up in the training and testing datasets.

4. **Target Variable Isolation (diameter):** All entries without a `diameter` have been removed. `Diameter (Y)` is the `target` variable and is required for the training model and to calculate the coefficient of determination (R^2_{test}) during out-of-sample testing.
5. **Exclusion of High-Cost Features (albedo):** `Albedo` represents a Group 3 characteristic and was explicitly dropped from the dataset. Removing this feature ensures the model does not rely on cost-prohibited data and maintains operational cost-effective analysis. This also prevents data leakage between `albedo` and the `target diameter`.
6. **Feature Subset Selection:** The model requires the `target diameter` and only features that fall into Group 1 (e.g., `semi-major axis a`, `eccentricity e`, `inclination i`) and Group 2 (`absolute magnitude H`).
7. **Feature Completeness:** Entries with missing values in the selected feature subsets were removed. This ensures the regression algorithms receive a complete feature matrix and that any observed variance in R^2_{test} is attributable to specific feature subsets and not fluctuations in sample size or missing data.
8. **Target Variable Distribution Scaling:** Scale diameter (`Y`) using a base-10 logarithmic formula. Small-bodies size distribution is non-linear. Scaling the `diameter (Y)` ensures that the model weights small and large small-bodies the same.
9. **Outlier Detection:** Because outliers represent a real small-body and not an entry error or out-of-scope feature, outliers will be reviewed but may not be removed just on the nature of being an outlier.
10. **Feature Independence:** Remove any highly correlated features to reduce the risk of improper feature ranking between the correlated features.

```
In [1]: import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import plotly.express as px
from sklearn.linear_model import LinearRegression

dataset_path = r"data/dataset-jpl-small-body-asteroids.csv"
dataset_cleaned_path = r"data/cleaned-dataset-jpl-small-body-asteroids.csv"

features = ['spkid', 'diameter', 'albedo', 'H', 'full_name', 'a', 'e', 'i', 'om', 'w', 'q', 'ad', 'per_y', 'data_arc']
drop_features = ['albedo']
x_restricted = ['a', 'e', 'i', 'om', 'w', 'q', 'ad']
x_full = ['H'] + x_restricted
y = ['diameter']
y_log = ['diameter_log']
```

```
full_model = y_log + x_full

df = pd.read_csv(dataset_path)
df.head()
```

Out[1]:

	spkid	diameter	albedo	H	full_name	a	e	i	om	w	q	ad	per_y	data_arc	condition_co
0	20000001	939.400	0.0900	3.34	1 Ceres (A801 AA)	2.766	0.0797	10.59	80.25	73.29	2.545	2.99	4.60	9520.0	
1	20000002	513.000	0.1550	4.12	2 Pallas (A802 FA)	2.770	0.2307	34.93	172.89	310.97	2.131	3.41	4.61	80828.0	
2	20000003	246.596	0.2140	5.19	3 Juno (A804 RA)	2.671	0.2557	12.99	169.81	247.90	1.988	3.35	4.37	80897.0	
3	20000004	522.770	0.4228	3.25	4 Vesta (A807 FA)	2.361	0.0902	7.14	103.70	151.47	2.148	2.57	3.63	25743.0	
4	20000005	106.699	0.2740	6.95	5 Astraea (A845 XA)	2.577	0.1876	5.36	141.45	359.38	2.093	3.06	4.14	65687.0	

1. Data Structure Integrity

All features are the correct data type. Discrete values like `spkid` (the small-body unique identifier) and `condition code` are integers (`int`) and all numerical measurements and values like `a` , `e` , and `i` are floats (`float64`).

In [2]: `df.info()`

```
<class 'pandas.DataFrame'>
RangeIndex: 139582 entries, 0 to 139581
Data columns (total 18 columns):
#   Column          Non-Null Count  Dtype
---  -
0   spkid            139582 non-null  int64
1   diameter         139582 non-null  float64
2   albedo           138459 non-null  float64
3   H                139203 non-null  float64
4   full_name        139582 non-null  str
5   a                139582 non-null  float64
6   e                139582 non-null  float64
7   i                139582 non-null  float64
8   om               139582 non-null  float64
9   w                139582 non-null  float64
10  q                139582 non-null  float64
11  ad               139582 non-null  float64
12  per_y            139582 non-null  float64
13  data_arc         139577 non-null  float64
14  condition_code   139582 non-null  int64
15  n_obs_used       139582 non-null  int64
16  n_del_obs_used   243 non-null     float64
17  n_dop_obs_used   243 non-null     float64
dtypes: float64(14), int64(3), str(1)
memory usage: 21.7 MB
```

2. Data Domain Validation

Removed data that is implausible. Since the data is not validated by an astrophysicist, the project will be limited to checking just the `diameter` (Y). Diameters are all valid because `(df['diameter'] <= 0).sum() = 0`.

```
In [3]: print(f"Number of invalid diameter records: {(df['diameter'] <= 0).sum()}")
```

Number of invalid diameter records: 0

3. Duplicate Data Removal

Removed rows with identical values in every column and where rows share the same `skpid`. Confirmed removed when `df.duplicated().sum() = 0` and `df.duplicated(subset=['skpid']).sum() = 0`.

```
In [4]: print("# of rows with identical values in every single column: ", df.duplicated().sum())  
print("# of rows with the same skipid", df.duplicated(subset=['skipid']).sum())
```

of rows with identical values in every single column: 0

of rows with the same skipid 0

4. Target Variable Isolation

All entries without a `diameter` (Y) have been removed. Confirmed because `.isnull().sum() = 0` for `diameter` (Y).

5. Exclusion of High-Cost Features

Drop `albedo` because it is a high-cost observation that will not be included in the model. Confirmed removed because feature does not show up in `.info()` or `.isnull().sum()`.

```
In [5]: df_clean = df.dropna(subset=y).drop(columns=drop_features)  
df_clean.isnull().sum()
```

```
Out[5]: skipid          0  
diameter          0  
H                379  
full_name        0  
a                0  
e                0  
i                0  
om              0  
w                0  
q                0  
ad              0  
per_y           0  
data_arc        5  
condition_code  0  
n_obs_used      0  
n_del_obs_used  139339  
n_dop_obs_used  139339  
dtype: int64
```

6. Feature Subset Selection

Removed all features that are not part of the full feature subset because they are metadata or not orbital characteristics used to determine `diameter` (Y).

```
In [6]: df_clean = df_clean[y + x_full]
df_clean.head()
```

```
Out[6]:
```

	diameter	H	a	e	i	om	w	q	ad
0	939.400	3.34	2.766	0.0797	10.59	80.25	73.29	2.545	2.99
1	513.000	4.12	2.770	0.2307	34.93	172.89	310.97	2.131	3.41
2	246.596	5.19	2.671	0.2557	12.99	169.81	247.90	1.988	3.35
3	522.770	3.25	2.361	0.0902	7.14	103.70	151.47	2.148	2.57
4	106.699	6.95	2.577	0.1876	5.36	141.45	359.38	2.093	3.06

7. Feature Completeness

Checked for any missing or null values using using `.isna().sum()`. If missing values, removed the entire entry using `.dropna()`. Since the only feature with missing values is `H` the entries can be deleted without review. This is because `H` is the feature difference in the hypothesis and is fundamental and required to test the model. For the same reasons, it is not appropriate to estimate this value.

```
In [7]: df_clean.isna().sum()
```

```
Out[7]: diameter      0
H                379
a                  0
e                  0
i                  0
om                 0
w                  0
q                  0
ad                 0
dtype: int64
```

```
In [8]: df_clean.dropna(subset=['H'], inplace=True)
df_clean.isna().sum()
```

```
Out[8]: diameter    0
H                0
a                0
e                0
i                0
om              0
w                0
q                0
ad              0
dtype: int64
```

8. Target Variable Distribution Scaling

Verified target diameter skew is greater than one (1). Since it was highly skewed, the diameter (Y) was transformed into a log base-10 scale, diameter_log , so the regression model does not improperly weight errors between small-body sizes. After the transformation, diameter_log was checked for errors in values and the entries were removed as necessary.

```
In [9]: df_clean['diameter'].describe()
```

```
Out[9]: count    139203.000000
mean         5.469118
std          9.318406
min          0.002500
25%          2.773000
50%          3.957000
75%          5.739000
max          939.400000
Name: diameter, dtype: float64
```

```
In [10]: print(df_clean['diameter'].skew())
```

```
25.930498594815756
```

```
In [11]: df_clean['diameter_log'] = np.log10(df_clean['diameter'])
print(f"Total infinity anomalies found: {np.isinf(df_clean['diameter_log']).sum()}")
print(f"Total na values found: {df_clean['diameter_log'].isna().sum()}")
df_clean['diameter_log'].describe()
```

Total infinity anomalies found: 0

Total na values found: 0

```
Out[11]: count    139203.000000
         mean      0.615454
         std       0.282694
         min       -2.602060
         25%       0.442950
         50%       0.597366
         75%       0.758836
         max       2.972851
         Name: diameter_log, dtype: float64
```

9. Outlier Detection

Calculated the lower and upper bounds from the `interquartile range` (`IQR`) and reviewed all outliers. No entries were determined to be errors so outliers were kept in the dataset because they represent a real small-body that is part of the population of data. The `diameter` was reviewed to compare to `diameter_log` but will not be used in the model.

```
In [12]: q1 = df_clean.quantile(0.25)
         q3 = df_clean.quantile(0.75)
         iqr = q3 - q1

         lower_bound = q1 - 1.5 * iqr
         upper_bound = q3 + 1.5 * iqr

         total_rows = len(df_clean)

         for col in df_clean.select_dtypes(include=['number']).columns:
             low_count = (df_clean[col] < lower_bound[col]).sum()
             high_count = (df_clean[col] > upper_bound[col]).sum()

             low_pct = (low_count / total_rows) * 100
             high_pct = (high_count / total_rows) * 100

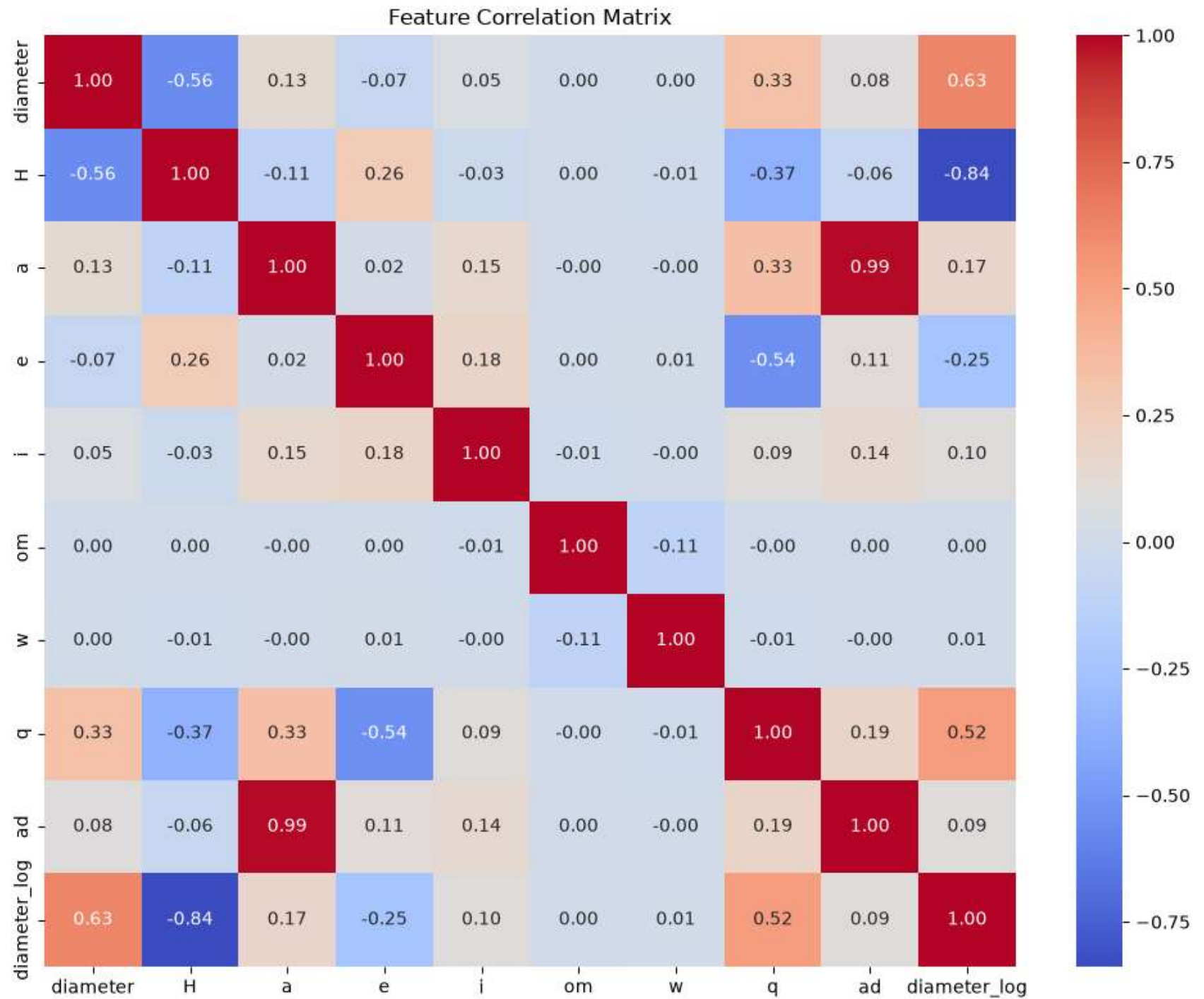
             print(f"{col} -> Lower Outliers: {low_count} ({low_pct:.2f}%) | Upper Outliers: {high_count} ({high_pct:.2f}%)")
```

```
diameter -> Lower Outliers: 0 (0.00%) | Upper Outliers: 9809 (7.05%)
H -> Lower Outliers: 3075 (2.21%) | Upper Outliers: 815 (0.59%)
a -> Lower Outliers: 579 (0.42%) | Upper Outliers: 3060 (2.20%)
e -> Lower Outliers: 0 (0.00%) | Upper Outliers: 1886 (1.35%)
i -> Lower Outliers: 0 (0.00%) | Upper Outliers: 4322 (3.10%)
om -> Lower Outliers: 0 (0.00%) | Upper Outliers: 0 (0.00%)
w -> Lower Outliers: 0 (0.00%) | Upper Outliers: 0 (0.00%)
q -> Lower Outliers: 1091 (0.78%) | Upper Outliers: 1983 (1.42%)
ad -> Lower Outliers: 449 (0.32%) | Upper Outliers: 3365 (2.42%)
diameter_log -> Lower Outliers: 840 (0.60%) | Upper Outliers: 3510 (2.52%)
```

10. Feature Independence

Created a heatmap to review highly correlated features. `a` and `ad` are highly correlated. Since `ad` is derived from `a` as `ad = a x (1 + e)` it will be removed so it does not get improperly weighted in the regression analysis. Features `om` and `w` have no feature-to-feature correlation or feature-to-target correlation. These will be kept since tree-based regression can factor these out if they are unnecessary.

```
In [13]: # create correlation matrix to see if any variables inherently correlate to diameter
plt.figure(figsize=(10,8))
sns.heatmap(df_clean.corr(numeric_only=True), annot=True, cmap='coolwarm', fmt=".2f")
plt.title("Feature Correlation Matrix")
plt.tight_layout()
plt.show()
```



```
In [14]: df_clean = df_clean.drop(columns=['ad'])
x_restricted.remove('ad')
x_full = ['H'] + x_restricted
full_model = y_log + x_full
print("Updated x_restricted:", x_restricted)
print("Updated x_full:      ", x_full)
print("Updated full_model:  ", full_model)
df_clean.head()
```

```
Updated x_restricted: ['a', 'e', 'i', 'om', 'w', 'q']
Updated x_full:      ['H', 'a', 'e', 'i', 'om', 'w', 'q']
Updated full_model:  ['diameter_log', 'H', 'a', 'e', 'i', 'om', 'w', 'q']
```

```
Out[14]:
```

	diameter	H	a	e	i	om	w	q	diameter_log
0	939.400	3.34	2.766	0.0797	10.59	80.25	73.29	2.545	2.972851
1	513.000	4.12	2.770	0.2307	34.93	172.89	310.97	2.131	2.710117
2	246.596	5.19	2.671	0.2557	12.99	169.81	247.90	1.988	2.391986
3	522.770	3.25	2.361	0.0902	7.14	103.70	151.47	2.148	2.718311
4	106.699	6.95	2.577	0.1876	5.36	141.45	359.38	2.093	2.028160

```
In [15]: # save cleaned data to csv
df_clean.to_csv(dataset_cleaned_path, index=False)
```

III. Analyze

Select analytical techniques, build models.

Data has been analyzed in the following ways:

1. **Individual Feature Boxplots:** Visualize distribution of individual features using `boxplots`.
2. **Target Normalization Check:** Visualize log transformation of `target diameter (Y)` using a `histogram`.
3. **Primary Signal Feature:** Visualize `absolute magnitude (H)` versus `target diameter (Y)` for correlation using a `scatterplot`.

4. **Orbital Characteristic Features:** Visualize features versus target diameter (Y), absolute magnitude (H), and other features for correlation and inclusion in model using 2D and 3D scatterplots with visual mapping.
5. **Excluded Features:** Determine features that do not require additional analysis.

1. Individual Feature Boxplots

Visually reviewed distribution of individual features. Several of the physical features (a, e, and i) are heavily skewed. This indicates linear regression models are not ideal and supports utilizing tree-based models. e (eccentricity) represents the orbit of a small-body with Earth having an $e \approx 0.017$. This means the small-body data set has a relative stable orbit similar to Earth while some outliers have less stable orbits. i represents inclination relative to Earth's (Earth's $i = 0^\circ$). This means the small-body dataset distribution of i has a similar inclination as Earth which makes sense given they are under the same orbital parameters as Earth. a represents the semi-major axis which is the average distance from the Sun. The distribution of a means that most of the small-bodies are in the main asteroid belt between Mars and Jupiter with some outliers as far out as Neptune or farther.

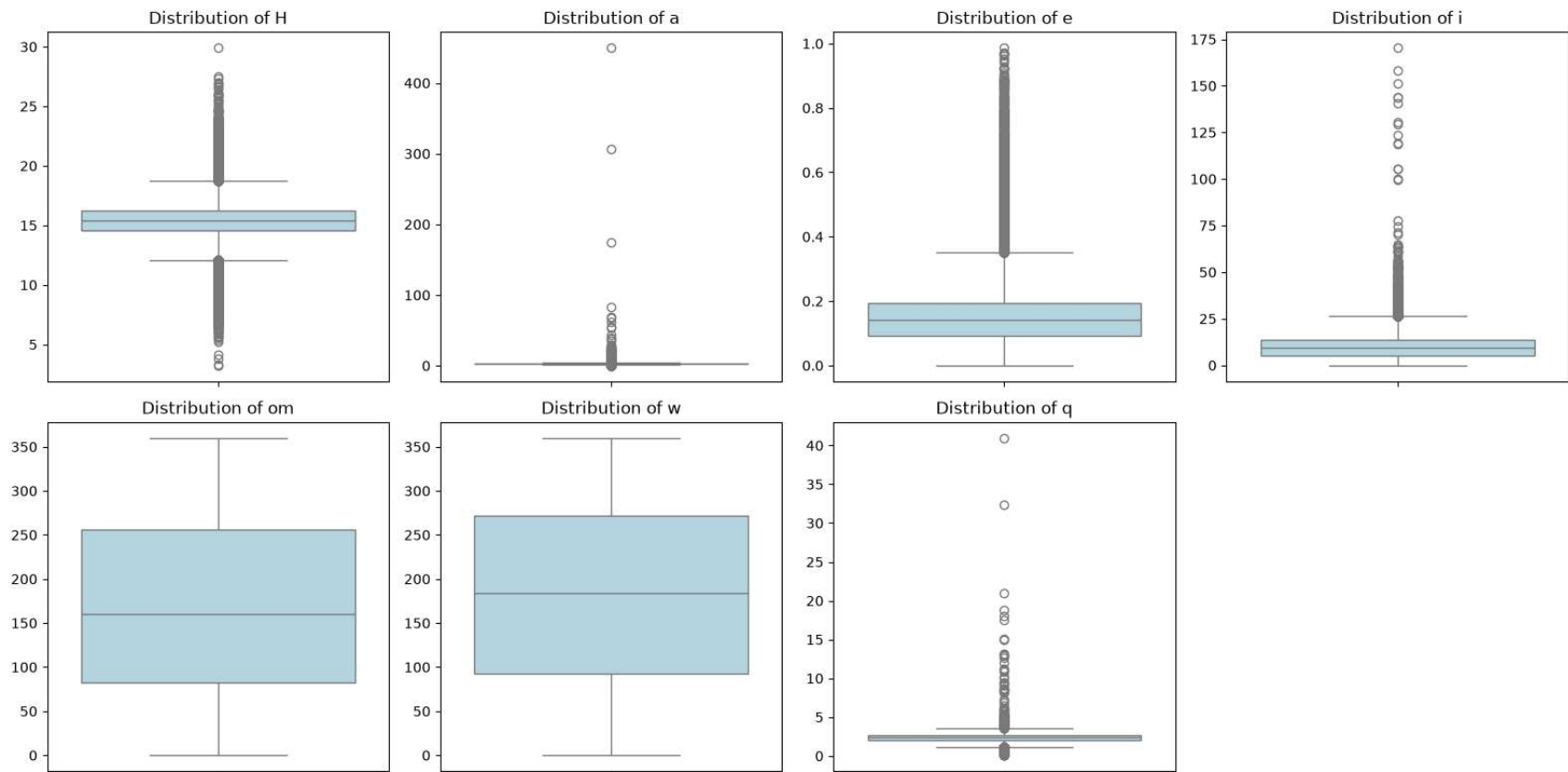
All features have acceptable distributions in context to small-bodies.

```
In [16]: fig, axes = plt.subplots(2, 4, figsize=(16, 8))
         axes = axes.flatten()

         for idx, col in enumerate(x_full):
             sns.boxplot(data=df_clean, y=col, ax=axes[idx], color='lightblue')
             axes[idx].set_title(f'Distribution of {col}')
             axes[idx].set_ylabel('')

         for i in range(len(x_full), len(axes)):
             fig.delaxes(axes[i])

         plt.tight_layout()
         plt.show()
```



2. Target Normalization Check

Compared the `diameter` to `diameter_log` to visualize skew. The raw `target diameter` is `power-law` skewed to the right. The `logarithmic` transformation decompresses the skew and better prepares the data for the `tree-based model` analysis.

```
In [17]: sns.set_theme(style="whitegrid")
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

sns.histplot(df_clean['diameter'], bins=50, kde=True, ax=axes[0], color='crimson')
axes[0].set_title('Raw Target: Asteroid Diameter')
axes[0].set_xlabel('Diameter (km)')
axes[0].set_ylabel('Frequency')

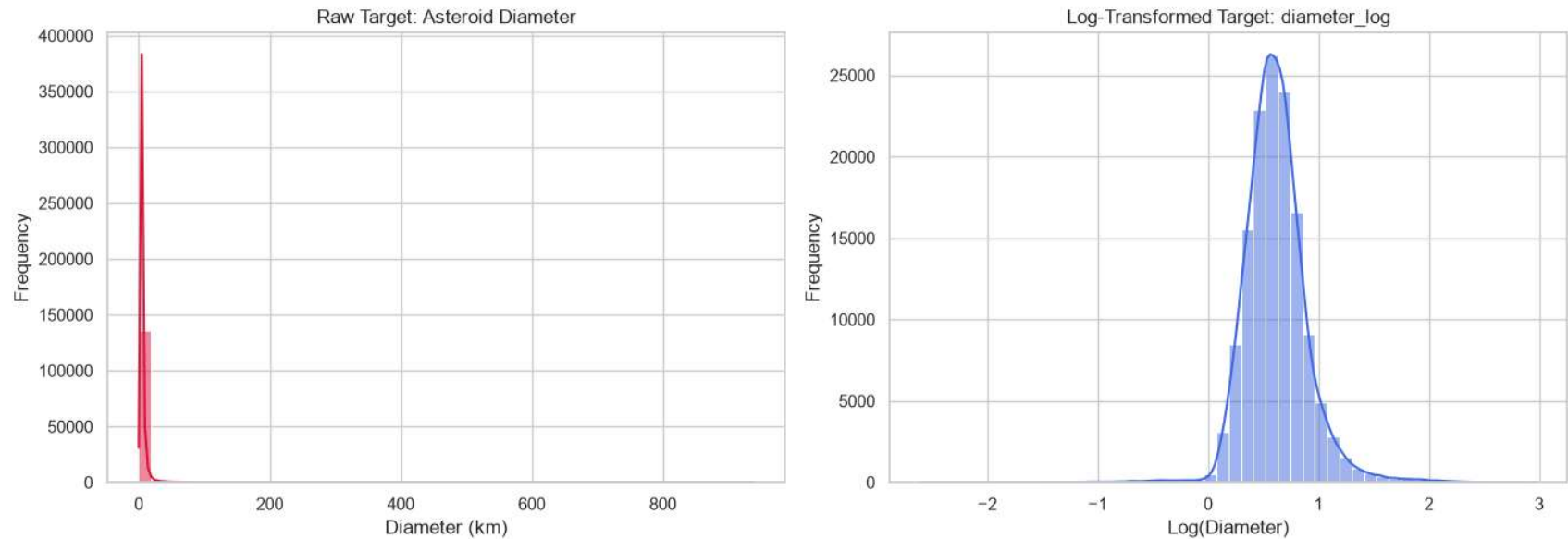
sns.histplot(df_clean['diameter_log'], bins=50, kde=True, ax=axes[1], color='royalblue')
axes[1].set_title('Log-Transformed Target: diameter_log')
```

```

axes[1].set_xlabel('Log(Diameter)')
axes[1].set_ylabel('Frequency')

plt.tight_layout()
plt.show()

```



3. Primary Signal Feature

H is the signal feature that differentiates the full and restricted feature subsets and is highly correlated to **diameter**. The scatterplot visually confirms this correlation and shows an inverse relationship between **H** and **diameter_log**. **Absolute magnitude (H)** represents **intrinsic brightness** of a small-body where a smaller **H** value means higher brightness. This trend shows that a small-body with a higher brightness (lower **H** value) tends to be larger and a small-body with a lower brightness (higher **H** value) tends to be smaller. This visualization supports the **peer-reviewed** research that **H** is highly correlated to **diameter** and important to predicting a small-body **diameter**.

```

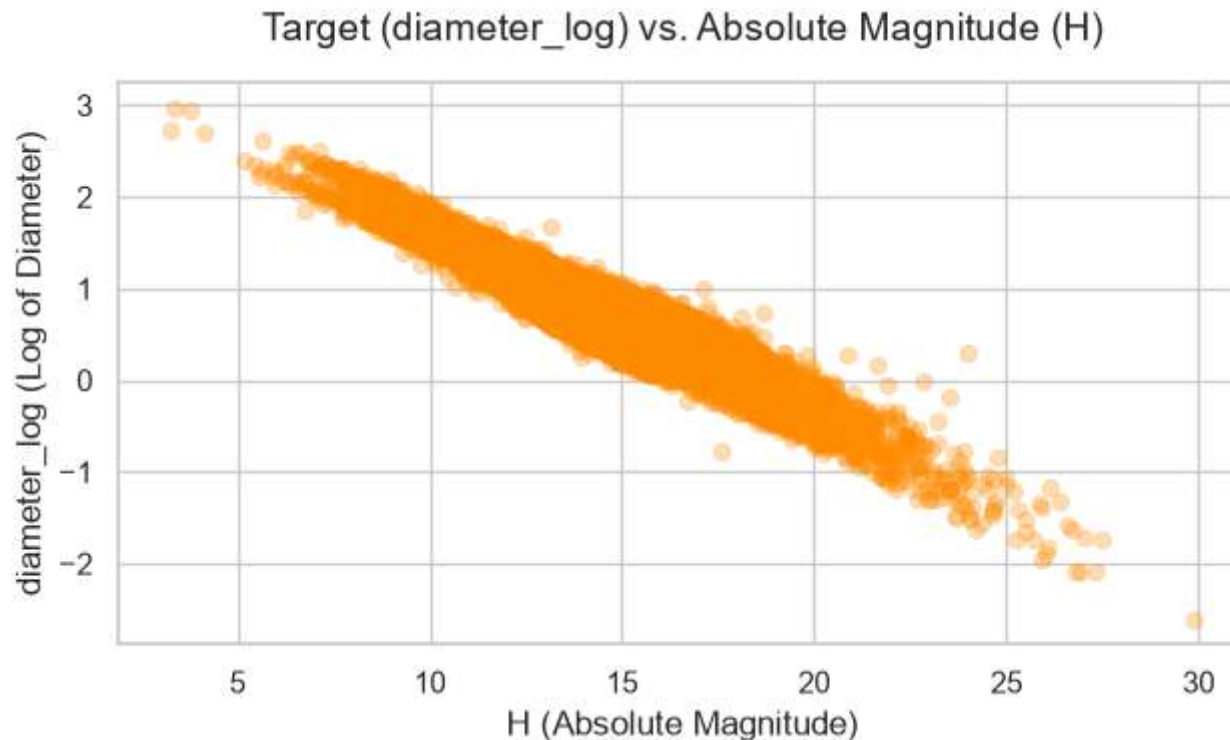
In [18]: sns.set_theme(style="whitegrid")
plt.figure(figsize=(6.5, 4))

sns.scatterplot(data=df_clean, x='H', y='diameter_log', alpha=0.3, color='darkorange', edgecolor=None)
plt.title('Target (diameter_log) vs. Absolute Magnitude (H)', fontsize=14, pad=15)
plt.xlabel('H (Absolute Magnitude)', fontsize=12)

```

```
plt.ylabel('diameter_log (Log of Diameter)', fontsize=12)

plt.tight_layout()
plt.show()
```



4. Orbital Characteristic Features

Fundamental Orbital Elements

Visualizing `a`, `e`, and `i` (independent orbital characteristics) vs. `diameter` to see if there is a correlation between these three features and `diameter`. The visualization reveals distinct clustering demonstrating that certain orbital groupings share diameter profiles. This provides strong preliminary evidence to support the `operational alternative hypothesis` that the features could increase accuracy in predicting `diameter` within an operationally acceptable range.

```
In [20]: df_viewport = df_clean[df_clean['a'] <= 10]

fig = px.scatter_3d(
```

```

df_viewport, x='a', y='e', z='i', color='diameter_log',
color_continuous_scale='viridis', opacity=0.5, title='Orbital Size, Shape, and Tilt vs Diameter',
labels={'a': 'a', 'e': 'e', 'i': 'i', 'diameter_log': 'diameter'},
)

fig.update_traces(marker=dict(size=2))
fig.update_scenes(xaxis_range=[0.5, 6.0], yaxis_range=[0, 0.5], zaxis_range=[0, 35])
fig.update_layout(margin=dict(l=0, r=0, b=0, t=40))
fig.show()

```

Absolute Magnitude (H) and Restricted Subset Fundamental Characteristics

Absolute Magnitude (H) versus Individual Restricted Subset Fundamental Characteristics

Visualize **a**, **e**, and **i** individually vs. **H** in terms of **diameter_log**. Visualization shows a clear indication that **H** is a primary indicator for predicting **diameter** but no distinct correlation exists between **H** and any of the individual features.

```

In [21]: sns.set_theme(style="whitegrid")
fig, axes = plt.subplots(1, 3, figsize=(18, 5))

sns.scatterplot(data=df_clean, x='a', y='H', hue='diameter_log', palette='viridis', alpha=0.5, s=8, ax=axes[0])
axes[0].set_title('Orbit Size (a) in AU vs. Absolute Magnitude (H)')
axes[0].set_xlabel('a')
axes[0].set_ylabel('H')
axes[0].set_xlim(0.5, 6.0)

sns.scatterplot(data=df_clean, x='e', y='H', hue='diameter_log', palette='viridis', alpha=0.5, s=8, ax=axes[1], legend=
axes[1].set_title('Orbit Shape (e) vs. Absolute Magnitude (H)')
axes[1].set_xlabel('e')
axes[1].set_ylabel('H')
axes[1].set_xlim(0, 0.6)

sns.scatterplot(data=df_clean, x='i', y='H', hue='diameter_log', palette='viridis', alpha=0.5, s=8, ax=axes[2], legend=
axes[2].set_title('Orbit Tilt in Degrees (i) vs. Absolute Magnitude (H)')
axes[2].set_xlabel('i')
axes[2].set_ylabel('H')
axes[2].set_xlim(0, 40)

axes[0].get_legend().remove()
plt.tight_layout()

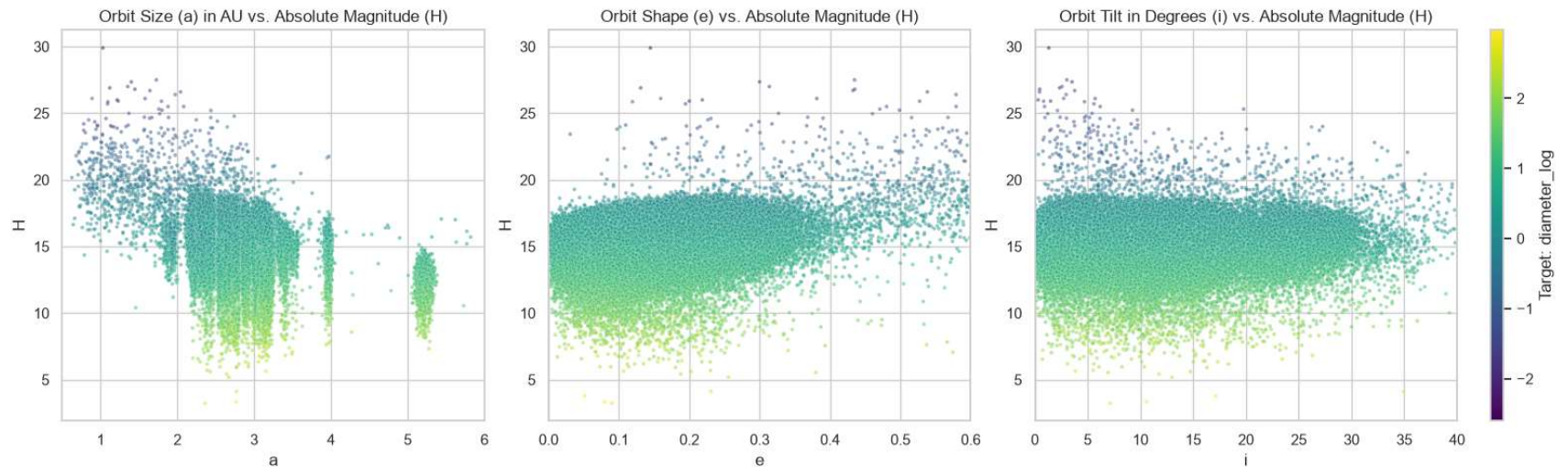
```

```

norm = plt.Normalize(df_clean['diameter_log'].min(), df_clean['diameter_log'].max())
sm = plt.cm.ScalarMappable(cmap="viridis", norm=norm)
sm.set_array([])
cbar = fig.colorbar(sm, ax=axes.tolist(), orientation='vertical', pad=0.02, aspect=30)
cbar.set_label('Target: diameter_log', fontsize=12)

plt.show()

```



Absolute Magnitude (H) in Relation to Restricted Subset Fundamental Characteristics

Visualizing **a**, **e**, and **i** (independent orbital characteristics) vs. **absolute magnitude (H)** to see if there is a correlation between these three features and **H**. The visualization reveals distinct clustering demonstrating that certain orbital groupings share magnitude profiles. This makes sense since **H** and **diameter** are closely related so the independent orbital characteristics should exhibit a nearly identical structural trend when mapped by **H** as they do when mapped by **diameter**.

```

In [22]: df_viewport = df_clean[df_clean['a'] <= 10]

fig = px.scatter_3d(
    df_viewport, x='a', y='e', z='i', color='H',
    color_continuous_scale='viridis', opacity=0.5, title='Orbital Size, Shape, and Tilt vs Diameter',
    labels={'a': 'a', 'e': 'e', 'i': 'i', 'H': 'H'},
)

fig.update_traces(marker=dict(size=2))
fig.update_scenes(xaxis_range=[0.5, 6.0], yaxis_range=[0, 0.5], zaxis_range=[0, 35])

```

```
fig.update_layout(margin=dict(l=0, r=0, b=0, t=40))
fig.show()
```

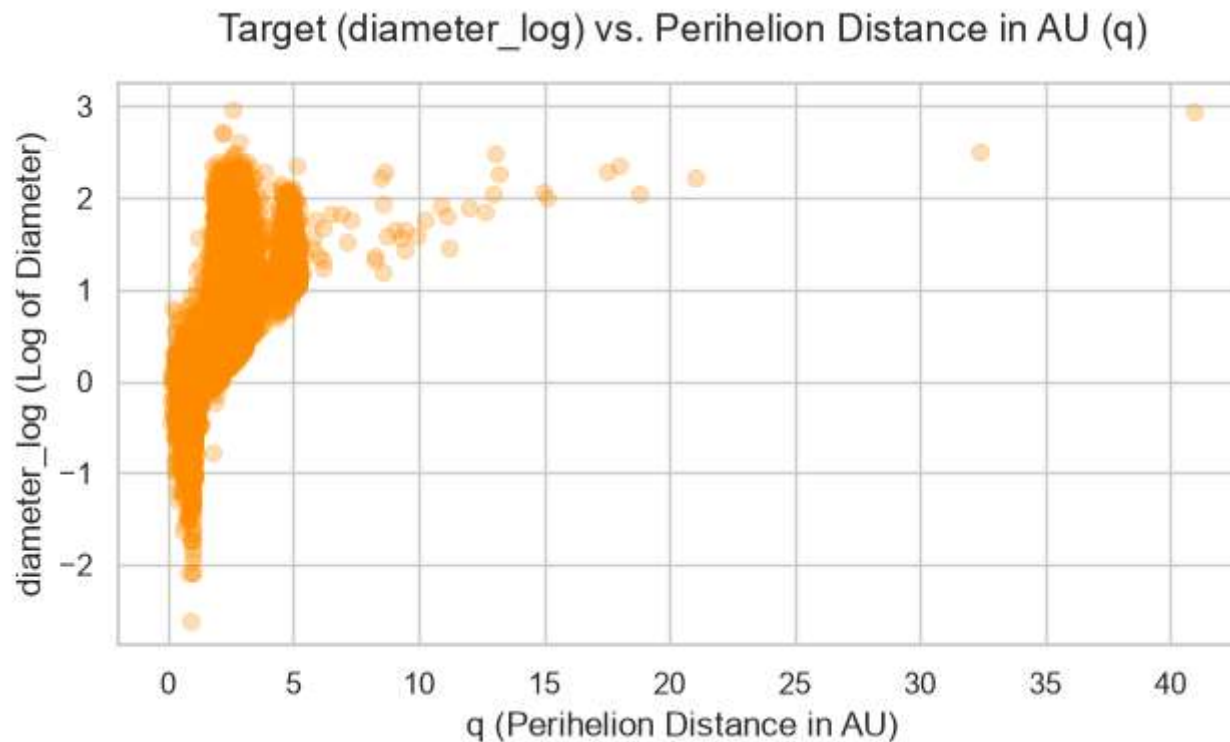
Derived Orbital Characteristics

q is a derived characteristic from a and e as $q = a(1-e)$. Visualizing q vs diameter_log shows a trend that the farther from the Sun a small-body gets, the larger it gets. IT has a strong linear slope specifcally at 1 AU up to 6 AU. Logically this does not make sense since the size of a small-body is not dictated by the distance from the Sun. There are small and large small-bodies near and far from the Sun. Since this is a bias in interpretation and not a flaw in the data, q can be kept in the dataset similar to om and w to be included or excluded in the final model determined at a later step.

```
In [23]: sns.set_theme(style="whitegrid")
plt.figure(figsize=(6.5, 4))

sns.scatterplot(data=df_clean, x='q', y='diameter_log', alpha=0.3, color='darkorange', edgecolor=None)
plt.title('Target (diameter_log) vs. Perihelion Distance in AU (q)', fontsize=14, pad=15)
plt.xlabel('q (Perihelion Distance in AU)', fontsize=12)
plt.ylabel('diameter_log (Log of Diameter)', fontsize=12)

plt.tight_layout()
plt.show()
```



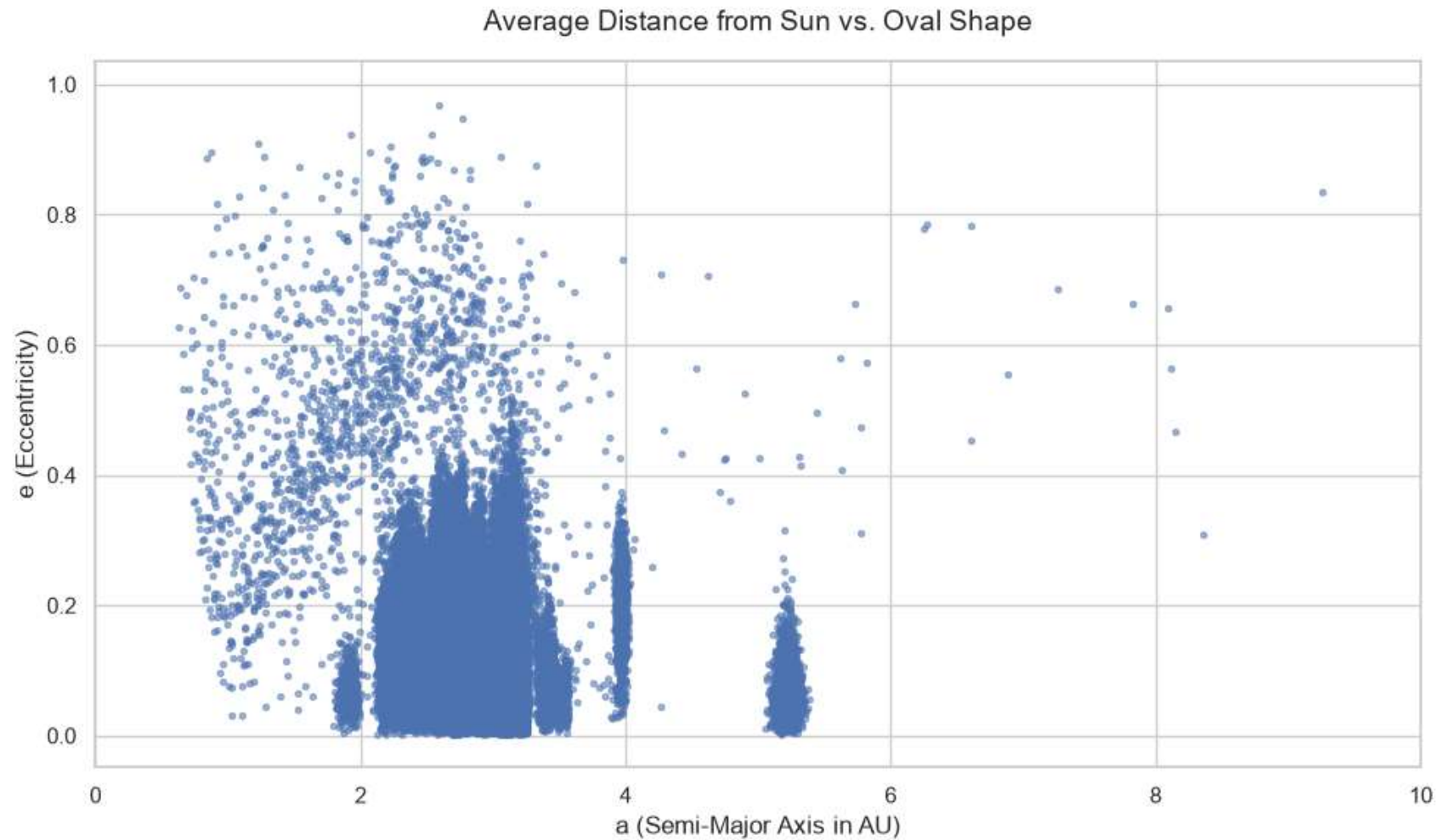
Semi-Major Axis v. Eccentricity

Visualizing `a` and `e` shows us the relationship between the average distance from the Sun and the orbital shape. The visualization does not show a relationship between `a` and `e` because there is not a `linear` relationship. The visualization does show that small-bodies tend to hold specific orbital distances with a major segment between 2 and 3.5 AU and another smaller segment around 5.2 AU. This is consistent with the main asteroid belt between Mars and Jupiter and a smaller one at Jupiter's orbit.

```
In [24]: sns.set_theme(style="whitegrid")
plt.figure(figsize=(10, 6))

scatter = plt.scatter(df_clean['a'], df_clean['e'], alpha=0.5, s=8)
plt.title('Average Distance from Sun vs. Oval Shape', fontsize=14, pad=15)
plt.xlabel('a (Semi-Major Axis in AU)', fontsize=12)
plt.ylabel('e (Eccentricity)', fontsize=12)
plt.xlim(0, 10)
```

```
plt.tight_layout()  
plt.show()
```



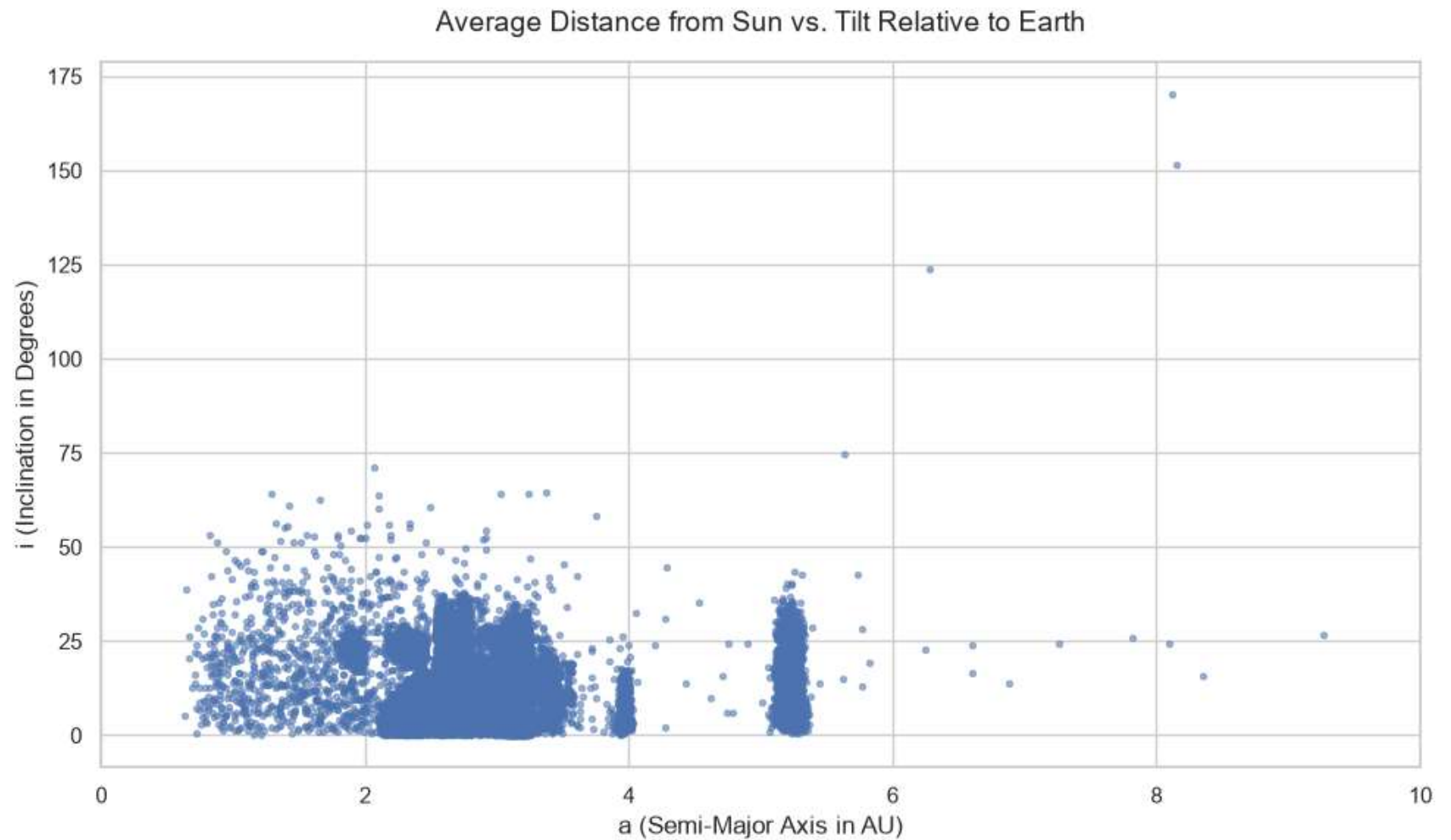
Semi-Major Axis v. Inclination

Visualizing `a` and `i` shows us the relationship between the average distance from the Sun and the inclination relative to Earth. The visualization does not show a relationship between `a` and `i` because there is not a linear relationship. The visualization does show that small-bodies throughout the inner solar system tend to have the same inclination as Earth.

```
In [25]: sns.set_theme(style="whitegrid")
plt.figure(figsize=(10, 6))

scatter = plt.scatter(df_clean['a'], df_clean['i'], alpha=0.5, s=8)
plt.title('Average Distance from Sun vs. Tilt Relative to Earth', fontsize=14, pad=15)
plt.xlabel('a (Semi-Major Axis in AU)', fontsize=12)
plt.ylabel('i (Inclination in Degrees)', fontsize=12)
plt.xlim(0, 10)

plt.tight_layout()
plt.show()
```



Eccentricity v. Inclination

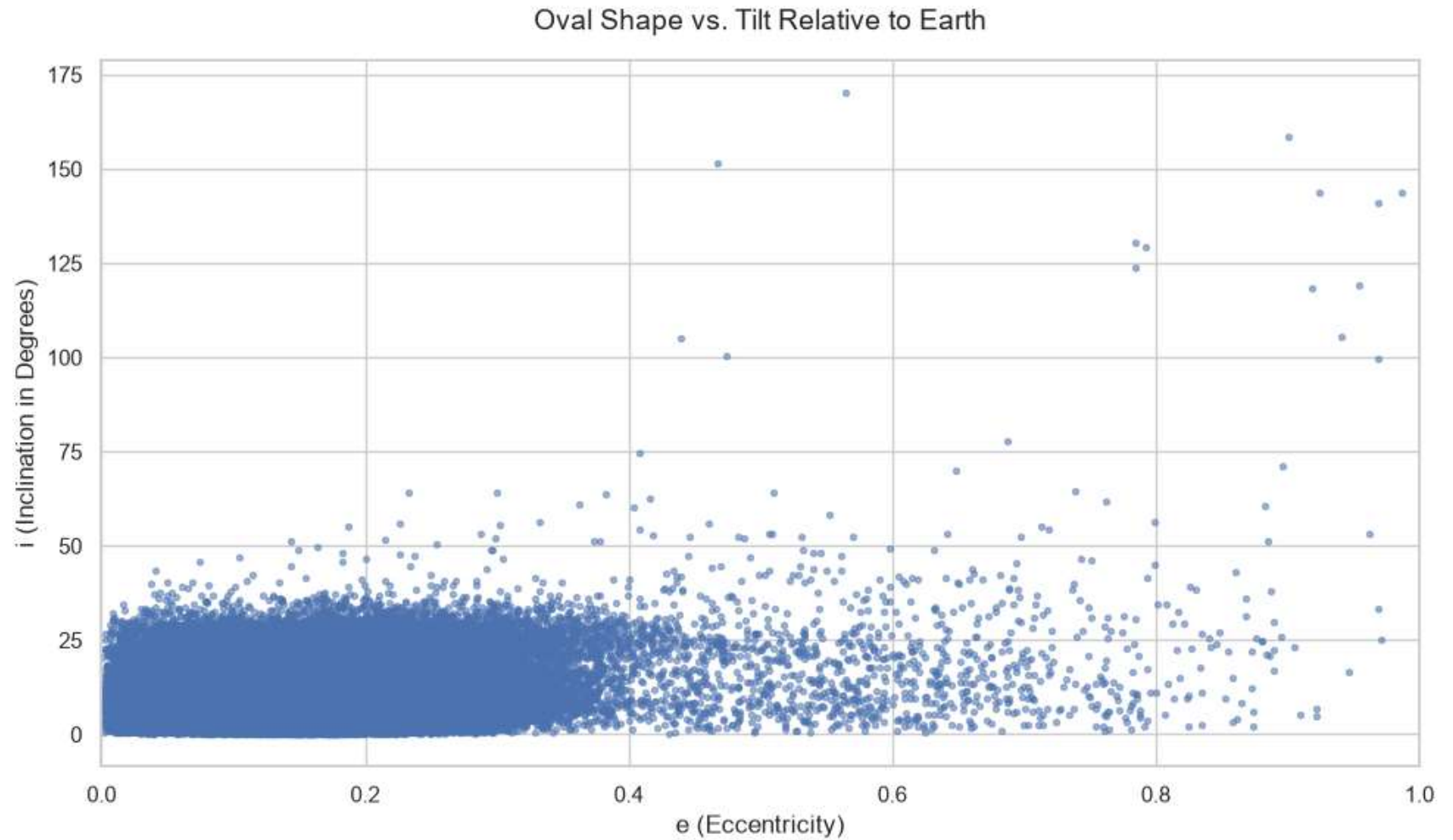
Visualizing `e` and `i` shows us the relationship between the orbital shape and the elliptical plane. The visualization does not show a correlation between `e` and `i` because there is not a linear relationship.

```
In [26]: sns.set_theme(style="whitegrid")
plt.figure(figsize=(10, 6))

scatter = plt.scatter(df_clean['e'], df_clean['i'], alpha=0.5, s=8)
plt.title('Oval Shape vs. Tilt Relative to Earth', fontsize=14, pad=15)
```

```
plt.xlabel('e (Eccentricity)', fontsize=12)
plt.ylabel('i (Inclination in Degrees)', fontsize=12)
plt.xlim(0, 1)

plt.tight_layout()
plt.show()
```



5. Excluded Features

`om` and `w` were excluded from visualization as the feature correlation matrix determined there was not a strong correlation between these features and other features or the target.

```
In [27]: df_clean = df_clean.drop(columns=['om', 'w'])
x_restricted.remove('om')
x_restricted.remove('w')
x_full = ['H'] + x_restricted
full_model = y_log + x_full
print("Updated x_restricted:", x_restricted)
print("Updated x_full:      ", x_full)
print("Updated full_model:  ", full_model)
df_clean.head()
```

```
Updated x_restricted: ['a', 'e', 'i', 'q']
Updated x_full:      ['H', 'a', 'e', 'i', 'q']
Updated full_model:  ['diameter_log', 'H', 'a', 'e', 'i', 'q']
```

```
Out[27]:
```

	diameter	H	a	e	i	q	diameter_log
0	939.400	3.34	2.766	0.0797	10.59	2.545	2.972851
1	513.000	4.12	2.770	0.2307	34.93	2.131	2.710117
2	246.596	5.19	2.671	0.2557	12.99	1.988	2.391986
3	522.770	3.25	2.361	0.0902	7.14	2.148	2.718311
4	106.699	6.95	2.577	0.1876	5.36	2.093	2.028160

```
In [28]: # save cleaned data to csv
df_clean.to_csv(dataset_cleaned_path, index=False)
```

IV. Report

1. Exploratory Data Analysis (EDA)

Preliminary analysis of the dataset shows that the key indicators of a small-body size (`target` `diameter` (`Y`)) are `absolute magnitude` (`H`), `semi-major axis` (`a`), `eccentricity` (`e`), and `inclination` (`i`). There is a strong `non-linear`

relationship between `diameter` and `H` indicating that `H` is a strong predictor of `diameter`. While there were no `linear` correlations between individual `a`, `e`, `i` features and `diameter` or `H`, comparison of `a`, `e`, and `i` mapped to `diameter` and `a`, `e`, and `i` mapped to `H` provided similar results. This provides strong evidence that `a`, `e`, and `i` could meet the operational threshold to predict `diameter` without the need for `H`. The `EDA` alone is not enough to conclude if the `null hypothesis` or the `alternative hypothesis` are correct, but there is a pattern and relationship between `a`, `e`, `i`, and `diameter`.

2. Feature Determination

Analysis of the features indicates that `a`, `e`, and `i` are strong candidates for the regression model because they have a `non-linear`, `structural` relationship to the target. The feature `q` is reserved for future determination for inclusion or exclusion in the regression model since it is derived from `a` and `e` which could introduce `multicollinearity` or `noise`. Features `om` and `w` can be excluded. While `om` and `w` are not derived from any other features, the lack of correlation to the `target` may introduce `noise` to the regression model that could lead to `overfitting`.

3. Model Selection

An analysis of the dataset and feature determination support using a `random forest` regression model to test the hypothesis. Small-bodies form dense structural clustering in the feature space and a highly skewed `target` distribution. Small-bodies also can be observed at great distances which creates mathematical `outliers`. Physical characteristics and derived characteristics may have no correlation to the size of a small-body or introduce `multicollinearity`. A `random forest` regression model uses piecewise step-function approximations to handle extreme `outliers` and `noise` of derived or irrelevant features. This is better than `linear` regression or other regression models that try to fit data into a straight line or curve. This will result in stronger models to increase accuracy to predict small-body sizes and to test the hypothesis.